ZCAS University

SMS4081 COMPUTER SCIENCE TEACHING METHODS 3

FINAL EXAMINATION

Date and Time: 25 NOVEMBER 2024 14:00 – 17:00 Hrs

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer Three (3) questions from this section.
3. This question paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1
Scenario

We and others have created ***case studies*** to engage students in the process of designing solutions to complex programming problems. Case studies involve learners in activities of expert programmers such as identifying decisions, justifying choices among alternatives, and evaluating the consequences of these choices. Case studies improve computer programming courses by emphasizing the process rather than the product of problem solving. Case studies emphasize the decisions encountered by the programmer and the criteria used to choose among alternatives.

Learners are engaged in the case study through questions that are interspersed in the presentation. Students can "*think along*" with the experts by making predictions, helping with parts of the solution, or analysing alternatives. More comprehensive questions ask students to modify the solution, apply the ideas to new problems, detect bugs in related problems, and reflect on their own methods for solving problems.

Required

By looking at use of ***case studies*** as an active teaching method, discuss how this method can be used to encourage the development of the ***Choice***, ***Collaboration***, ***Creativity***, ***Critical Thinking***, ***Cooperation*** and **Communication** skills, in other words the 6Cs, among learners. Marks shall be awarded for answers that relate to examples cited in the scenario in this question.

(Total: 40 marks)

SECTION B FOLLOWS ON THE NEXT PAGE

SECTION B: Attempt any THREE questions in this section

Question 2
Scenario

Preparing a child for the world that doesn't yet exist is not an easy task for any teacher. Step back and look at that picture from a broad perspective. What are the critical 21st century skills every student needs to survive and succeed in our world? What abilities and traits will serve them in a time that's changing and developing so rapidly?

No pupil in the history of education is like today's modern learner. This is a complex, energetic, and tech-savvy individual. They want to be challenged and inspired in their learning. They want to collaborate and work with their peers. They want to incorporate the technology they love into their classroom experiences as much as they can. In short, they have just as high a set of expectations of their educators as their educators have of them.

The Global Digital Citizen Foundation has given hundreds of presentations to educators and administrators in several countries over the years. Each time they have spoken, they have asked educators what they feel are the most important 21st-century skills needed by students above all others. The answers that they have received most often are narrowed down to the following:

- Problem solving
- Creativity
- Analytic thinking
- Collaboration
- Communication
- Ethics, action, and accountability

Required

Explain why each of these skills is important and how they are likely to help 21st century learners to succeed.

(20 marks)

SECTION B CONTINUES ON THE NEXT PAGE

SECTION B CONTINUED

Question 3

The 21st century skills that we've just gone through in Question 2 are considered the essentials for our students' future success. These all won by a landslide vote from the education professionals who have given their thoughts on why they feel these skills are most important.

Required:

Discuss what these 21st century skills look like in the classroom, and how you would measure them?

[20 marks]

Question 4

In this question, you are required to solve a teaching problem using the following steps:

i) Identify a PROBLEM you encounter in your teaching.

ii) Identify possible REASONS for the problem.

iii) Explore STRATEGIES to address the problem.

The following are some of the possible problems you may encounter in your teaching:

a) Students come late to class

b) Students don't demonstrate critical thinking.

c) Students lack interest or motivation.

d) Students cannot apply what they have been taught.

e) Students respond to course content and classroom dynamics in emotional and unproductive ways.

Required:

Identify possible REASONS for each problem, bearing in mind that there may be many other possible reasons, and explore possible STRATEGIES to address each problem.

[20 marks]

SECTION B CONTINUES ON THE NEXT PAGE

SECTION B CONTINUED

Question 5

In a "***Flipped Classroom***", students' **initial exposure to content** is shifted **outside of the classroom** via readings, instructional videos, individual or collaborative activities, or a combination of these. Then **during class**, rather than lecturing, all or a significant portion of the time is used for practice, application exercises, discussion-based activities, team-based learning, or other **active learning techniques**. Some preliminary assessment, such as an online quiz or brief assignment, may be used to gauge student understanding and tailor instructional plans prior to class.

Required:

Discuss five (5) benefits of this approach to students and five (5) benefits to teachers. *Remember that all teaching strategies (e.g., flipped classroom vs. traditional lecture) have pros and cons depending on context and how they are used. One is not necessarily inherently superior to another. Effectiveness for learning ultimately depends on how strategies are implemented.*

[20 marks]

END OF FINAL EXAM – NOVEMBER 2024